

Run 02 | Fear of failure

India

IV only

Target Fear of failure prevents starting a business; binary 1 vs. 0

Sample 27,276 respondents **Coverage** 1 country; 2015–2022

Positive cases 12,009 (44.03%) **Case time** 1m 02s

Train / val. / test 18,549 / 3,273 / 5,454 (68% / 12% / 20%, stratified)

SCT predictors (12): Self-efficacy, Opportunity perception, Knowing entrepreneurs, Entrepreneurial intention, Media coverage, Business exit, Business angel, Established business owner, Good career choice, Ease of starting a business, High status, Social enterprise.

Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 25.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.7310). Highest F1: XGBoost (0.6605). The all-negative accuracy baseline is 55.97% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.7310	0.6834	59.48	52.33	89.42	0.6602	2s
XGBoost	0.7269	0.6747	61.68	54.14	84.67	0.6605	1s
CatBoost	0.7268	0.6733	63.09	55.73	78.55	0.6520	21s
H2O GBM	0.7225	0.6701	62.72	55.24	80.76	0.6561	5s
Logistic Regression	0.6813	0.6153	53.32	48.43	92.92	0.6367	<1s
Ridge	0.6811	0.6157	53.41	48.47	92.09	0.6351	11s
Elastic Net	0.6811	0.6170	53.32	48.41	92.00	0.6344	7s
Lasso	0.6808	0.6172	53.41	48.46	92.04	0.6350	7s
Decision Tree	0.6753	0.6012	52.62	48.06	94.38	0.6369	<1s
Naive Bayes	0.6435	0.5660	52.82	48.06	88.96	0.6241	6s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Good career choice	193.20	8.46
2	Media coverage	146.71	6.42
3	Opportunity perception	143.05	6.26
4	High status	132.91	5.82
5	Social enterprise	126.81	5.55
6	Entrepreneurial intention	117.90	5.16
7	Self-efficacy	96.70	4.23
8	Ease of starting a business	94.26	4.13

Share is the percentage of total forest impurity importance across all 49 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Education	48.94	2.14
2	Investment environment	43.26	1.89
3	Health	41.93	1.84

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 4 batch, 29 September 2026; 02_obj4_india_iv_fearfail_2015_2022. Scores and importance are taken directly from saved CSV outputs.